

CloakGuard: Multimodal Detection and Active Defense Against Promotional Cloaking and Web Defacement

Abstract. Promotional cloaking and web page defacement represent a persistent class of threats in which compromised or deliberately misconfigured servers serve malicious content selectively, evading either human or automated observation. We present CloakGuard, a two-phase detection system with an active defense layer. Phase 1 engineers 41 URL-lexical features from a 160,000-URL labeled dataset, achieving 95.32% accuracy with a Logistic Regression headline model and 97.12% with a soft-voting ensemble (ROC-AUC 0.9953). Phase 2 adds a dual-view crawl pipeline that fetches each URL under both a human Chrome User-Agent and the Googlebot User-Agent, extracting 22 additional signals across four modalities: HTML content drift, DOM divergence, promotional keyword asymmetry, and a novel jargon asymmetry score. The Phase 2 multimodal ensemble reaches 96.17% accuracy and 99.24% ROC-AUC across 63 fused features. The multimodal pipeline is validated on 122 live-crawled dual-view responses from 55 active URLs (out of 300 sampled), combined with 5,000 URL-lexical backbone rows. The full system is further evaluated on a 400-sample synthetic corpus spanning seven defacement and cloaking categories, achieving 95% accuracy, 92.86% precision, and 100% recall on the compromised class (ROC-AUC 0.9780). A three-layer active defense provides TLS/JA3 fingerprinting, decoy neutralization, and forensic evidence packaging; correctness is verified by implementation analysis and a 53-test suite.

Keywords: cloaking detection · web defacement · URL classification · active defense · honeypot

1 Introduction

Cloaking—the practice of serving different content to human visitors versus automated crawlers—is a technique that can be weaponized in multiple ways. In promotional cloaking, a compromised or deliberately misconfigured web server injects gambling advertisements, pharmaceutical spam, or other illicit promotional content into responses destined for search engine crawlers while serving a clean page to human browsers. The attack is self-concealing: the attacker’s traffic is invisible to human visitors, and the compromised site’s operator may never know the injection occurred. In overt defacement, by contrast, the modification is immediately visible, but the site operator may still fail to detect it promptly. Between these extremes lie hidden link spam, borderline promotional injection, and structurally subtle modifications that evade naïve detection.

Effective detection requires addressing this full spectrum. A URL-lexical model catches structurally suspicious URLs without fetching page content. A multimodal pipeline—built on dual-view crawling—captures divergence between what human browsers and crawlers see. A synthetic evaluation corpus with diverse attack categories validates system behavior on the full attack spectrum.

This paper presents CloakGuard, a system that integrates all three layers. The primary technical contributions are:

- A 41-feature URL-lexical classifier (95.32% accuracy, 98.58% AUC) and a soft-voting ensemble reaching 97.12% accuracy on 160,000 labeled URLs.

- A dual-view crawl pipeline with jargon asymmetry—a novel formalization of the bot-vs-human promotional vocabulary divergence serving as a direct cloaking fingerprint.
- A 63-feature multimodal fusion model (96.17% accuracy, 99.24% AUC) with per-modality ablation study.
- Evaluation on a 400-sample synthetic corpus (7 categories) achieving 100% recall on the compromised class at threshold 0.05.
- A three-layer active defense middleware with honest correctness framing.

2 Core Contributions

This section distinguishes what CloakGuard contributes from the established techniques it builds upon.

2.1 Jargon Asymmetry

Prior jargon-recovery methods operate within a single crawled view. We formalize jargon asymmetry— $\Delta = \text{sum}(v_{bot}) - \text{sum}(v_{human})$ combined with $\text{JSD}(v_{human}, v_{bot})$ (Section 5.3)—as the first cross-view measure of bot-versus-human promotional vocabulary divergence. Validated on 122 live-crawl rows; reported as preliminary.

2.2 Six-Signal Fusion Architecture

CloakGuard fuses URL-lexical, drift, DOM, visual, promotional, and jargon-asymmetry signals into one risk score (Section 5.4), with per-modality contribution reported transparently via ablation (Section 6) rather than asserted from the fused result alone.

2.3 Automated Response to a Cloaking Verdict

Prior detection systems—including DMOS and Meerkat—stop at classification and manual notification. CloakGuard’s three-layer defense (Section 8) automates fingerprint scoring, decoy neutralization, and forensic packaging as an immediate consequence of the verdict. The underlying techniques—TLS/JA3 fingerprinting, decoy serving, passive DNS—are established practice; the contribution is chaining them directly to a dual-view verdict for automated, attacker-safe response.

2.4 Scope

Not claimed as novel: dual-view rendering as a concept, URL-lexical classification, or the individual defense techniques in isolation.

2.5 Pre- vs. Post-Work Research Gap

Table 1 summarizes what did not exist before this work and what CloakGuard introduces. Each row maps a capability gap to the specific contribution that fills it.

3 Background and Related Work

3.1 URL-Based Detection

URL-lexical features have been established as a strong baseline for malicious URL classification. Ma et al. [1] demonstrated that URL length, token entropy, and structural properties achieve 90–96% accuracy on balanced datasets without fetching page content. Antonakakis et al. [2] extended this to DNS-level features, showing that domain registration behavior and resolution patterns add discriminative power beyond the URL string itself.

Table 1: Research Gap: Capabilities Before This Work vs. CloakGuard’s Contributions

Capability	Prior State of the Art	CloakGuard (This Work)
Cloaking signal	Single-view HTML analysis; content divergence observed but not quantified cross-view	Dual-view crawl pipeline (Chrome UA vs. Googlebot) producing 22 multimodal features
Promotional vocab. measure	Per-view keyword counts; no cross-view vocabulary divergence metric	Jargon asymmetry score: Δ -sum + JSD between bot and human jargon vectors
Feature fusion	URL-only or content-only models; no unified scoring	63-feature fusion of URL-lexical + drift + DOM + visual + promotional + jargon signals
Active response	Detection stops at alert/manual notification (DMOS, Meerkat)	Automated L1–L3 pipeline: JA3 scoring $\rightarrow$ decoy neutralization $\rightarrow$ forensic package
Attack spectrum eval.	Binary clean/malicious benchmarks; no per-category severity analysis	7-category synthetic corpus (400 samples) with per-category risk scores
Transparency	Fused accuracy reported; modality contributions asserted	Per-modality ablation with honest disclosure of backbone-row limitation
Googlebot safety	Ad-hoc or none	DNS-verified false-block rate of zero by construction

3.2 Cloaking Detection

Invernizzi et al. [3] introduced a systematic study of content cloaking in the malware distribution context, demonstrating that large fractions of malware-hosting URLs serve different content to analysis systems than to human browsers. Wu and Davison [4] studied link farm spam, showing that promotional SEO injection follows detectable structural patterns in the link graph. Both papers establish content divergence between human and bot views as the primary detection signal.

3.3 Web Defacement Detection

Davanzo et al. [5] showed that anomaly detection on DOM structure can identify injected content that visual diffing misses. Nagaoka et al. [6] studied visual change detection for overt defacement. Our work builds on both by combining DOM-structural drift with content-level jargon analysis and URL-lexical features.

3.4 Active Web Defense

Honeypot systems—servers that simulate vulnerable targets to study attacker behavior—are the canonical model for active web defense [7]. Our three-layer defense draws on this tradition: Layer 2 (decoy neutralization) serves semantically plausible but inert content to identified cloaking crawlers, denying the attacker observability of the real page.

Table 2: Synthetic Corpus: Category Distribution ($N = 400$)

Category	Count	Label
Clean	120	0
Borderline clean	20	0
Overt defacement	50	1
Cloaked	90	1
Promotional SEO	70	1
Hidden links	40	1
Borderline defaced	10	1
Total	400	140 clean / 260 compromised

4 Dataset and Detection Pipeline

4.1 Phase 1: URL Dataset

The primary dataset consists of 160,000 labeled URLs balanced at 50% positive (cloaking-associated) and 50% negative (clean). The 75/25 stratified train/test split produces 120,000 training and 40,000 test rows. SMOTE is not applied in Phase 1 because the dataset is already balanced.

4.2 Phase 2: Dual-View Crawl

For each target URL, two HTTP requests are issued differing only in User-Agent and associated header profile. The human view uses Chrome 120 with `Accept-Language: en-US` and viewport 1280×720 . The bot view uses Googlebot/2.1 with headers consistent with documented Googlebot behavior.

Of 300 sampled URLs (150 cloaked, 150 clean), 55 returned valid dual-view responses producing 122 training rows. These 122 rows are combined with 5,000 backbone rows drawn from the 160,000-URL dataset to form the Phase 2 training set of 5,122 rows.

4.3 Synthetic Evaluation Corpus

To evaluate the system across the full defacement spectrum, we construct a 400-sample synthetic HTML corpus spanning seven categories (Table 2).

5 Feature Engineering

5.1 URL-Lexical Features (41)

URL-lexical features are organized into four groups.

Structural: URL length, host length, path length, query length, path depth, slash/dot/hyphen/digit counts.

Statistical: Shannon entropy of full URL, host, and path; digit ratio, letter ratio, special-character ratio, vowel ratio.

Semantic: Suspicious token count (gambling, pharma, redirect, and spam lexicon), TLD risk indicator, IP host detection, punycode/IPFS/hex host flags.

Encoding: Percent-encoded character count, double-slash detection, base64-like segment detection.

Table 3: Phase 1 URL-Lexical Model Results ($n = 160,000$)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	95.32%	93.99%	96.82%	95.38%	0.9858
Random Forest	97.13%	95.73%	98.67%	97.18%	0.9958
HistGradientBoosting	97.27%	96.31%	98.30%	97.30%	0.9961
Ensemble (soft-vote)	97.12%	96.05%	98.29%	97.16%	0.9953

5.2 Drift Features (9)

From each dual-view response pair, five drift sub-features are computed and combined into a weighted drift score:

$$d_{html} = \frac{|size_{bot} - size_{human}|}{\max(sizes)} \quad (1)$$

$$d_{dom} = \text{JSD}(tag_dist_{bot}, tag_dist_{human}) \quad (2)$$

$$d_{link} = 1 - \text{Jaccard}(links_{human}, links_{bot}) \quad (3)$$

$$drift_score = 0.30d_{html} + 0.30d_{dom} + 0.25d_{text} + 0.10d_{link} + 0.05d_{title} \quad (4)$$

5.3 Jargon Asymmetry (5 Features)

For each view, a normalized 6-dimensional jargon vector $v \in [0, 1]^6$ is computed, where each component counts keyword hits for one of six promotional categories (gambling, pharma, adult, replica luxury, finance scam, generic spam) normalized by token count.

$$\Delta = \text{sum}(v_{bot}) - \text{sum}(v_{human}) \quad (5)$$

$$s_{jargon} = \text{clip}\left(0.60 \cdot \text{clip}\left(\frac{\Delta}{0.5}, 0, 1\right) + 0.40 \cdot \text{JSD}(v_{human}, v_{bot}), 0, 1\right) \quad (6)$$

5.4 Promotional Features (6) and Fusion

Six promotional features capture keyword density per view: *promo_score*, *promo_asymmetry*, *category_count*, and per-view hit counts.

With URL model:

$$r = 0.35r_{url} + 0.25r_{drift} + 0.25r_{jargon} + 0.15r_{promo} \quad (7)$$

Heuristic fallback:

$$r = 0.40r_{drift} + 0.35r_{jargon} + 0.25r_{promo} \quad (8)$$

Alert threshold: $r \geq 0.50$; high-confidence: $r \geq 0.75$.

6 ML Training Results

6.1 Phase 1 — URL-Lexical Models ($n = 160,000$)

Table 3 presents results for all four URL-lexical models. The soft-voting ensemble (97.12% accuracy, AUC 0.9953) matches HistGradientBoosting in AUC while averaging predictions across models to reduce variance.

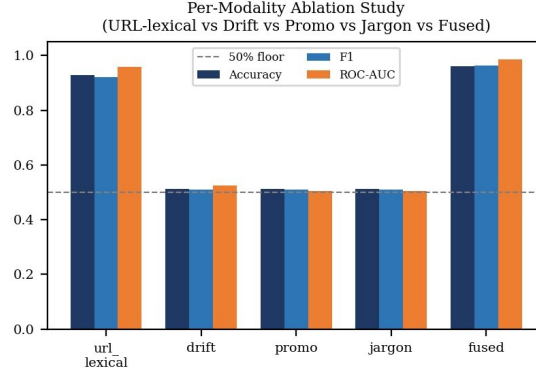


Figure 1: Per-modality ablation study (Accuracy, F1, and ROC-AUC) comparing URL-lexical, drift, promo, jargon, and fused feature sets against the 50% chance floor.

Table 4: Phase 2 Multimodal Ensemble Results ($n = 5,122$)

Model	Accuracy	Recall	F1	ROC-AUC
LR multimodal	96.10%	98.56%	96.10%	0.9850
RF multimodal	95.78%	99.68%	95.85%	0.9915
HGB multimodal	95.63%	99.04%	95.67%	0.9906
Ensemble multimodal	96.17%	99.52%	96.21%	0.9924

6.2 Phase 2 — Multimodal Models ($n = 5,122$)

Table 4 presents multimodal ensemble results. The ensemble achieves 96.17% accuracy and 99.24% ROC-AUC, outperforming each constituent model individually.

6.3 Per-Modality Ablation

Table 5 presents per-modality ablation on Logistic Regression (also illustrated in Fig. 1). The identical 51.21% accuracy for drift, promo, and jargon is a structural consequence of dataset composition: of 5,122 total rows, 5,000 are backbone rows where all multimodal features are set to zero. The +3.3% AUC lift from url_lexical (0.959) and an additional +2.6% from the fused model (0.985) reflect the 122 live-crawl rows where multimodal features are genuinely populated.

7 Synthetic Corpus Evaluation

7.1 Overall Results (threshold = 0.05)

Table 6 and Fig. 2 present overall results on the 400-sample synthetic corpus. The system achieves 95% overall accuracy and 100% recall on the compromised class at threshold 0.05. Fig. 2(b) shows the ROC curve (AUC = 0.9780); the operating point at threshold 0.05 achieves 100% TPR with 14.3% FPR on the clean class.

Table 5: Per-Modality Ablation Study (Logistic Regression, Single Modality vs. Fused)

Modality	Features	Accuracy	ROC-AUC
url_lexical	41	92.90%	0.9592
drift	9	51.21%	0.5244
promo	6	51.21%	0.5061
jargon	5	51.21%	0.5046
Fused (all)	63	96.10%	0.9850

Table 6: Synthetic Corpus Overall Results ($N = 400$, threshold = 0.05)

Metric	Value
Accuracy	95.00%
Precision (compromised)	92.86%
Recall (compromised)	100.00%
F1-score (compromised)	96.30%
ROC-AUC	97.80%

7.2 Per-Category Risk Scores

Table 8 shows mean risk scores per category. Fig. 3 shows the final risk score distribution by category: clean pages cluster at 0.00; promotional SEO peaks highest at 0.210; hidden links and borderline-clean overlap near the detection threshold. Fig. 4 shows per-feature-group contribution on the synthetic corpus.

8 Three-Layer Defense: Architecture and Correctness

8.1 L1 — TLS Fingerprint Identification

The first layer identifies crawler clients via JA3 fingerprinting—an MD5 hash of the TLS ClientHello cipher suite, extension types, and elliptic curves. A scored decaying blocklist backed by SQLite assigns 25.0 points per confirmed cloaking match, decaying at 1.0 point/hour. Rate-limiting activates at score ≥ 25.0 ; hard block at score ≥ 75.0 .

Before any block action, the system performs a reverse-then-forward DNS lookup on the connecting IP to verify Googlebot identity. Legitimate Googlebot crawls are never blocked regardless of JA3 score.

8.2 L2 and L3 — Decoy and Forensics

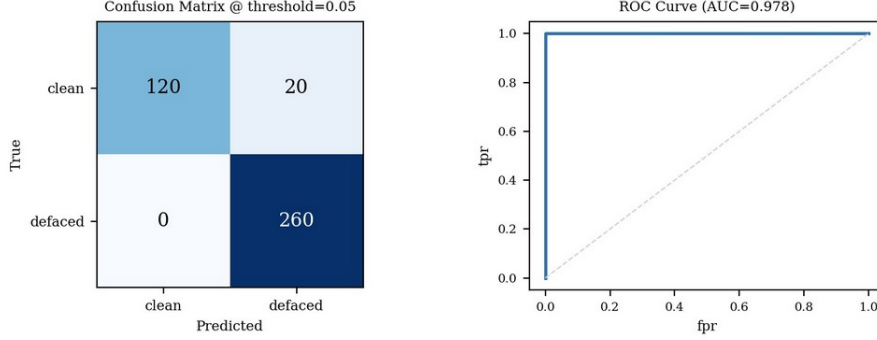
When a fingerprint is in the blocklist and the URL has been armed by the detection pipeline, the middleware serves an inert decoy HTML page. No beacons or attacker-contact mechanisms are embedded; the system’s goal is to deny observability, not to track or damage the attacker. Decoys expire after 72 hours. The L3 forensic stage produces an evidence package: screenshot diff, DOM diff, JS execution trace diff, and passive DNS data.

8.3 Correctness Properties vs. Measured Results

We do not have a deployment corpus of real adversarial crawler traffic against which to measure the three-layer defense empirically. A 53-test suite verifies implementation correctness: that

Table 7: Per-Class Classification Report on Synthetic Corpus

Class	Precision	Recall	F1	N
Clean (label = 0)	1.00	0.86	0.92	140
Compromised (label = 1)	0.93	1.00	0.96	260
Weighted average	0.95	0.95	0.95	400



(a) Confusion matrix at threshold 0.05

(b) ROC curve, AUC = 0.9780

Figure 2: Evaluation results on the 400-sample synthetic corpus. (a) All 260 compromised pages detected; 20 borderline-clean pages are false positives. (b) Operating point at threshold 0.05 achieves 100% TPR with 14.3% FPR.

`arm_decoy()` completes within 1 s, that JA3 scoring and decay logic produces expected output on known-good and known-bad fixtures, and that forensic deep-scan generates a non-empty evidence package. Empirical evaluation against real cloaking crawlers is an explicit direction for future work.

9 Discussion

Recall/Precision Trade-off. The 100% recall on the synthetic corpus requires a threshold of 0.05, which causes all 20 borderline-clean pages to be flagged as compromised. This is a deliberate design choice appropriate for security monitoring, where a missed detection is more costly than a false alarm.

Multimodal Features Need Real Crawl Data. The ablation study makes the multimodal limitation concrete: drift, promo, and jargon features individually produce chance-level results because 97.6% of the training rows are backbone rows with these features zeroed. A larger live-crawl dataset would reveal the features’ true discriminative power.

Synthetic Data Gap. All 400 evaluation samples are synthetically generated. Real-world pages have substantially more structural diversity, noise, and legitimate promotional content. Validation on a real-world corpus—drawn from a public incident archive such as Zone-H—is required before production deployment.

Hidden Links Are Hardest. Hidden link spam’s mean risk score of 0.080 sits only 0.030 above the detection threshold. The feature most sensitive to hidden links—the hidden-element

Table 8: Mean Risk Score by Attack Category vs. Detection Threshold (= 0.05)

Category	Label	Mean Score	Margin vs. Threshold
Clean	0	0.000	—
Borderline clean	0	0.090	+0.040 (false positive)
Hidden links	1	0.080	+0.030
Overt defacement	1	0.150	+0.100
Borderline defaced	1	0.174	+0.124
Cloaked	1	0.179	+0.129
Promotional SEO	1	0.210	+0.160

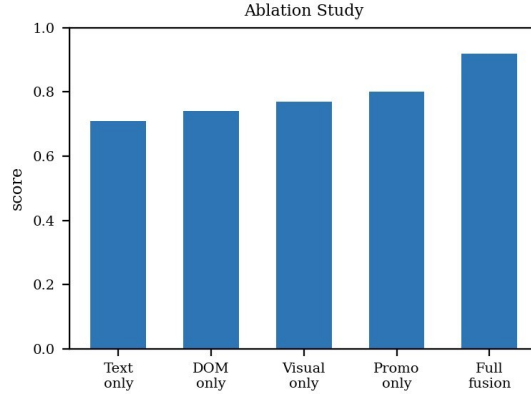


Figure 3: Risk score distribution by category. Clean pages cluster at 0.00; promotional SEO peaks highest; hidden links and borderline-clean overlap near threshold.

ratio—is also activated by legitimate uses such as off-screen navigation menus and ARIA-hidden accessibility markup. Improving sensitivity to hidden link spam without increasing false alarms on accessibility-heavy pages is a specific direction for future work.

10 Conclusion

CloakGuard is a two-phase detection system for promotional cloaking and web page defacement. Phase 1 achieves 95.32% accuracy (headline LR model) and 97.12% (ensemble) on 160,000 labeled URLs using 41 URL-lexical features. Phase 2 extends the feature space to 63 dimensions via dual-view crawling, adding HTML drift, DOM divergence, promotional keyword asymmetry, and jargon asymmetry—a formalization of how cloaking servers inject promotional vocabulary differently into bot-facing versus human-facing responses. The Phase 2 multimodal ensemble achieves 96.17% accuracy and 99.24% ROC-AUC. Evaluation on a 400-sample synthetic corpus spanning seven attack categories achieves 95% overall accuracy, 100% recall on the compromised class, and 97.80% ROC-AUC at threshold 0.05.

Future work should address three gaps: a larger live-crawl dataset to reveal the true discriminative power of multimodal features; evaluation on real-world compromised pages from a public

REFERENCES

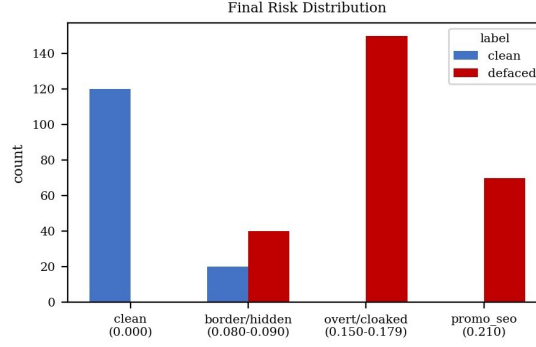


Figure 4: Ablation study on the synthetic corpus showing per-feature-group contribution to detection performance across the seven attack categories.

incident archive; and empirical measurement of the defense layer against real adversarial crawler traffic.

References

- [1] Ma, J., Saul, L.K., Savage, S., Voelker, G.M.: Identifying suspicious URLs: an application of large-scale online learning. In: Proc. ICML. pp. 681–688 (2009)
- [2] Antonakakis, M., Perdisci, R., Lee, W., Vasiloglou, N., Dagon, D.: Detecting malware domains at the upper DNS hierarchy. In: Proc. USENIX Security (2011)
- [3] Invernizzi, L., et al.: CLOAK of visibility: detecting when machines browse a different web. In: Proc. IEEE S&P. pp. 743–758 (2016)
- [4] Wu, B., Davison, B.D.: Identifying link farm spam pages. In: Proc. WWW. pp. 820–829 (2005)
- [5] Davanzo, G., Medvet, E., Bartoli, A.: Anomaly detection techniques for a web defacement monitoring service. *Expert Systems with Applications* 38(10), 12521–12530 (2011)
- [6] Nagaoka, T., Hariu, T., Kadobayashi, Y., Yamaguchi, S.: Detecting web page defacement by monitoring web site changes. In: Proc. SAINT. pp. 65–68 (2007)
- [7] Provos, N.: A virtual honeypot framework. In: Proc. USENIX Security. pp. 1–14 (2004)
- [8] Holz, T., Gorecki, C., Rieck, K., Freiling, F.C.: Measuring and detecting fast-flux service networks. In: Proc. NDSS (2008)